

Gigabyte updates entry level Mobos

Gigabyte has updated its entry level boards with features like 4-phase power design, Ultra Durable 2 technology, DualBIOS, 3TB+ HDD Support and Solid Capacitors for CPU VRM

Tegra 3 to roll out in MWC?

NVIDIA's Tegra 3 is rumoured to roll out at the upcoming Mobile World Congress in Barcelona

Motherboard test

While the GD65 and GD55 are higher end boards which look quite similar except for the extra two SATA ports on GD65, the C45 is a low end P67 board with only one PCIe x16 slot. GD65 and GD55 boards have dedicated power, reset and OC Genie buttons which are absent on the C45.

The ECS Black Series P67H2-A is one of the most attractive looking boards from amongst all that we tested. The colour combination is quite pleasing and it has sturdy heat sinks. The piece de resistance, as far as this board is concerned, is the presence of a Lucid Hydra Logix core, which will allow you to connect two cards from different manufacturers and run them together. Which means you can combine the NVIDIA GTX580 with an AMD HD6970!

The Intel DP67BG reference board was well designed. It has a skull towards the bottom edge whose eyes glow when the system is switched on. There is a nice representation of status of all the attached components in the form of green LED indicators on the Intel board.

The H67 boards

We received five H67 board from ASUS, ECS, Biostar and Gigabyte. All the boards except Gigabyte H67A-UD3H came in the microATX form factor. Although some boards came with two PCIe x16 slots, the limitation of the H67 chipset prevents them from running a dual - GPU SLI or Crossfire setup. All boards except ECS H67H2-M had their SATA ports pointing

upward, which can sometimes get in the way of the graphic card if it is long. Heatsinks on H67H2-M were the best among the lot. Gigabyte H67A-UD3H had strong heatsinks, but they were flexible, something we did not like. Both the ASUS boards sported IDE slots, which are too old now and seemed like a waste of space. ECS board impressed us with dedicated power and reset buttons as well as two digit POST code LED on board. H67 being a board that allows you to utilise the internal graphics of the Sandy Bridge processor, do come with as many as four display ports namely HDMI, Display-Port, DVI and VGA. All boards except ASUS P8H67-M came with all the four display ports.

Performance and verdict

ECS H67H2-M scored 11000 plus in overall PC Mark Vantage score followed by ASUS P8H67-M. Cinebench rendering was impressive in P8H67-M as well as Gigabyte H67A-UD3H. Absence of USB 3.0 ports on the P8H67-M board was quite strange, as all the other boards had them. Real life tests including data transfer and gaming were very close as well.

As far as H67 boards were concerned, we found the Gigabyte H67A-UD3H and ASUS P8H67-M-EVO to be good in terms of performance and features, but the economical pricing of Biostar TH67XE under ₹7,000 ensured that it won the **Best Buy**. ASUS P8H67-M was disappointing in terms of both features and performance. Despite ASUS having the UEFI BIOS, it still scored lower in features when compared to the rest of the boards.

For P67 boards, Gigabyte P67A-UD7 scored the most in Cinebench Rendering with 19471, which was about 400 points over its nearest competitor - ECS P67H2-A. WinRAR compression / decompression was fastest in MSI P67A-C45 board followed by ASUS P8P67 PRO. The difference was miniscule among all the boards. Gaming scores were almost similar with most of the boards giving great playable rates.

Gigabyte P67A-UD7 won the **Best Performer** as its performance score edged past all other com-

HOW WE TESTED

The new motherboards have the LGA 1155 sockets and it is not backward compatible. We used the fastest Sandy Bridge processor - the Intel Core i7-2600K running at 3.4GHz for all the boards. Corsair's XMS3 (1600MHz) 4GB DDR3 RAM was used along with a Sparkle GTX 465 graphics card. The memory was clocked at 1333MHz running at 9-9-9-24. During the test, Windows 7 Ultimate edition ran off a Western Digital Velociraptor 300GB hard drive. All the components were powered by Corsair HX620W power supply unit.

We divided the motherboards based on the P67 / H67 chipset. We ran PCMark Vantage to test the performance across various usage scenarios. That was followed by the ScienceMark Memory Bandwidth test. CineBench R10, an independent tool based on the popular 'Cinema 4D' 3D modelling program was used to test just the processor and memory. Another way to test the processor was by using WinRAR 3.9 built-in benchmark tool. Coming to the real world tests, we tested the time taken to encode a .VOB video file into the DivX format using DivX Converter 7.0. That was followed by a single large file and assorted 4GB file transfer test to simulate sequential and assorted from one partition to the other. As most motherboards come with a USB 3.0 port, we also ran data transfer tests on USB 3.0 ports. In order to see how these boards performed in the graphics department, we tried running Crysis Warhead benchmark, Resident Evil 5 benchmark and Company of Heroes : Tales of Valour benchmarks. For H67 boards, we tested the integrated graphics prowess by running the same games at lower resolutions off the display ports on these boards after removing the discrete graphics card.

peting boards' scores. ECS P67H2-A was awarded the **Editor's Pick** as its overall score was very close to the Best Performer and for being the only motherboard to support Lucid Hydra Logix core.

We felt that MSI P67A-C45 was not really upto the mark. Maybe MSI is targetting the budget segment, but Intel's DP67BG was quite superior to it in both features and performance. It was the **Best Buy**. We would have liked to see the UEFI BIOS on more boards though. 



ECS Black Series P67H2-A